

Battery Unbalance In Substation Batteries

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Abstract—This paper presents a final project for the Digital Signal Processing (DSP) course, addressing the application of statistical analysis for monitoring and analyzing lead-acid battery banks in electric energy substations. The primary goal is to identify imbalances and anomalies that could compromise the provision of uninterrupted power by analyzing variables such as individual cell voltage, internal resistance, temperature, and electrolyte density. To this end, data acquisition, processing, and statistical analysis techniques will be employed, enabling the creation of histograms and the computation of metrics such as mean, variance, and standard deviation. The expected outcomes aim to provide a more accurate diagnosis of battery bank imbalance, facilitating capacity testing to prevent failures.

Index Terms—Digital Signal Processing, Battery Monitoring, Lead-Acid Batteries, Statistical Analysis, Battery Imbalance.

I. INTRODUCTION

In critical electrical installations, such as electric energy substations (EES), power supply interruptions are entirely unacceptable, as they may compromise the operation of essential systems and cause significant financial losses. To ensure continuous power supply, even in situations of primary grid failure, the use of Uninterruptible Power Supply (UPS) systems is crucial.

Under power outage conditions, batteries supply critical systems such as control, protection, and telecommunication equipment. In other words, these batteries are primarily used in emergency situations and are therefore rarely subjected to frequent charge and discharge cycles. In this context, stationary lead-acid batteries are widely used due to their high cost-effectiveness compared to more modern technologies like lithium-ion batteries. Lead-acid batteries are particularly well-suited for applications where volume and cycle life are not critical, but where safety, reliability, and ease of maintenance are essential [1].

In this context, substations employ lead-acid battery banks (BB) connected in series, forming battery assemblies with voltages ranging from 48 V to 250 V, depending on the project specifications, load requirements, and standards adopted by the utility company.

When a failure occurs in the primary power source, the battery banks are connected directly to the DC loads present in the substation, without conversions, for more reliability. However, the use of multiple series-connected cells may result in a voltage imbalance among individual batteries.

This voltage imbalance can reduce overall lifespan. To address this issue, this paper presents a solution that uses

a specific electronic device aimed at mitigating imbalance effects. To contextualize the problem, a statistical analysis is conducted using histogram-based evaluation.

This paper analyzes lead-acid battery banks in electric substations through the application of statistical techniques and Digital Signal Processing (DSP) methods. The objective is to identify imbalance patterns and propose a corrective solution based on power electronics. The dataset originates from substations containing 60 series-connected elements, which exhibited test failures due to voltage drops in imbalanced cells.

Furthermore, to improve safety in a controlled environment, we tested an electronic converter to solve imbalance issues using an emulated setup with fewer cells. Where parameters such as electrolyte density, temperature, and voltage are monitored. Statistical tools are utilized to evaluate the collected data, and DSP techniques are employed to process the signals.

Through DSP techniques, it is possible to acquire and process data such as voltage, temperature, and electrolyte density, allowing for the creation of histograms and graphs that illustrate the distribution of these parameters. Furthermore, essential statistical metrics such as mean and standard deviation are calculated.

The remainder of this paper is organized as follows: Section II presents the discharge test performed at a utility company to illustrate the imbalance problem. Section III describes the experimental procedure using a reduced set of elements.

II. BATTERY IMBALANCE IN SUBSTATIONS

The imbalance occurs when one or more elements in the battery bank exhibit a voltage significantly different from the average group voltage. This issue arises because the power converter responsible for charging the batteries controls only the total voltage of the BB, without individually monitoring each element. This approach contrasts with more modern systems, such as Battery Management Systems (BMS) used in lithium-ion batteries, which ensure individual voltage equalization.

In addition to identifying imbalances, it is crucial to highlight that when a significant deviation from the mean is detected, a specialized power converter is employed to balance the individual cells. This converter works to correct any voltage discrepancies between the cells, ensuring that all battery modules remain balanced. As a result, premature cell degradation is prevented, thereby extending the overall lifespan of the battery bank. This approach not only reduces the need

for early disposal but also contributes to cost savings and enhances environmental sustainability.

A. Discharge Test

Among the periodic tests prescribed for the maintenance of substation battery elements, capacity tests are particularly significant.

The batteries are subjected to a discharge test at their nominal capacity. The procedure involves a 10% discharge of the capacity. The test is considered complete when the battery bank is able to reach its full capacity. When "failures" occur, it means that the element reaches the minimum voltage threshold (1.75 V). When this happens, the element is removed until the third element fails. At this point, the test is considered concluded, and the battery bank is considered "failed."

The primary objective of this test is to validate the original functionality of the system, ensuring that element imbalance does not lead to failures during routine maintenance checks, as defined by standards such as NBR 14199 [2] and NBR 14205 [3], which regulate battery bank assessments in electrical substations.

Table I presents a data excerpt from a capacity discharge test performed as part of routine monitoring by a utility company. These tests are crucial for verifying the ability of the battery system to supply energy in the event of a power outage in an Electrical Substation System (ESS).

Analysis of the data reveals that certain battery elements show voltage deviations from the majority of the cells at the very beginning of the discharge process, indicating a lower State of Charge (SOC) compared to the rest. This imbalance suggests early degradation or improper charging of those specific cells.

TABLE I: Excerpt from a capacity discharge test of a 60-celled lead-acid battery bank showing elements with voltages below 1.8 V (orange) and 1.75 V (red)

n.	Flut. (V)	0.5h (V)	1h (V)	1.5h (V)	2h (V)	2.5h (V)	3h (V)	4h (V)
1	2.27	1.99	1.98	1.96	1.94	1.92	1.90	1.87
2	2.28	1.99	1.98	1.93	1.91	1.89	1.88	1.78
3	2.27	1.98	1.96	1.94	1.93	1.90	1.88	1.80
4	2.27	1.99	1.97	1.95	1.94	1.92	1.87	1.83
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
54	2.26	1.99	1.97	1.95	1.94	1.92	1.89	1.82
55	2.26	1.98	1.97	1.94	1.93	1.91	1.84	1.77
56	2.26	1.98	1.97	1.94	1.93	1.90	1.84	1.75
57	2.26	1.98	1.97	1.94	1.93	1.90	1.83	1.72
58	2.26	1.98	1.97	1.94	1.93	1.91	1.84	1.73
59	2.26	1.98	1.97	1.93	1.91	1.88	1.85	1.74
60	2.26	1.97	1.96	1.94	1.92	1.90	1.87	1.70

The discharge test may be repeated with up to two fewer elements; however, the entire charging, float, and equalization process must be repeated, which is time-consuming and may span several days. If, after repeating the test, the battery bank still fails to meet the required capacity, the entire bank is deemed unsuitable and must be replaced.

The proposed solution aims to address one of the primary causes of such failures: voltage imbalance among the elements of the battery bank.

Given that this study emphasizes the application of statistical techniques in voltage signal measurements, a discharge test dataset from the aforementioned company is extracted and analyzed. Three histograms are generated, each corresponding to a different stage of the discharge procedure:

- Initial stage: Immediately after the beginning of discharge (fluctuation voltage 2.2 V/e);
- Intermediate stage: thirty minutes of the discharge begins;
- Final stage: The third cell presents voltage below 1.75 V).

These histograms are used to evaluate the evolution of the voltage distribution among the cells and to clearly identify any imbalances visually.

B. Statistical analysis

Several figures of merit are extracted from the data for analysis, such as the mean, mathematically defined as:

$$\mu = \frac{1}{N} \sum_{i=0}^{N-1} x_i \quad (1)$$

Additionally, the standard deviation (σ) is computed, which is a measure of how far the signal fluctuates from the mean, defined as:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=0}^{N-1} (x_i - \mu)^2 \quad (2)$$

Since samples from 60 cells were acquired, it is visually difficult to analyze this in tables. The histogram helps to quickly and visually show the number of cells that are unbalanced.

Figure 1. presents the histograms corresponding to the different stages of operation described earlier. The graph generated is characterized by a categorical histogram, where each category represents a voltage range of the bank.

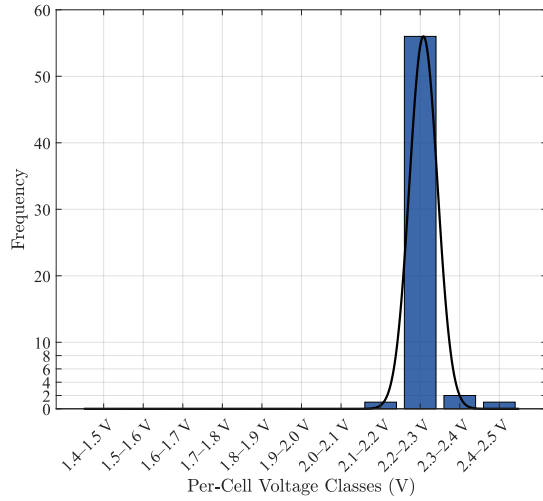
C. Discussion of the results

In the first stage, the beginning of the test is observed, as illustrated in Figure 1 (a). The float voltage is 2.2 V, and it is verified that most elements exhibit average voltages within expected values, with three elements showing voltages above the average and one below. At this phase, the existence of an unbalanced cell cannot yet be confirmed.

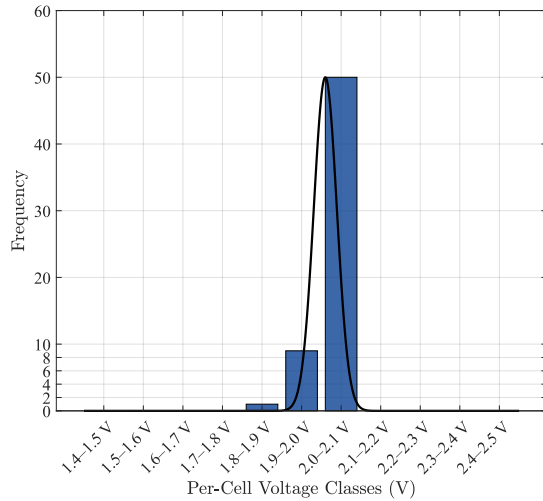
In the second stage as illustrated in Figure 1 (b), after thirty minutes of discharge, some elements are observed not to exhibit voltages below 1.75 V. However, some cells already show a voltage significantly different from the others, indicating that their state of charge (SoC) is considerably lower than the rest, thus characterizing a pronounced imbalance.

In the third stage, the imbalance becomes more evident, with one element showing a considerable difference relative to the others, totaling four unbalanced elements.

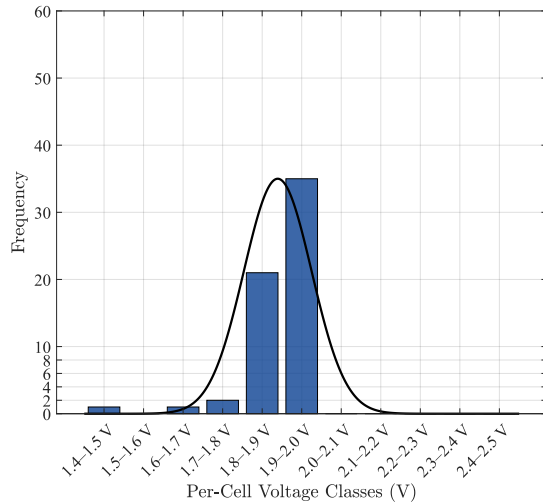
It is important to highlight the element with 1.4 V, as the test is performed until the elements reach 1.75 V. This is because



(a) Float stage: $\mu = 2.2582$, $\sigma = 0.0354$



(b) After 30 minutes: $\mu = 2.0100$, $\sigma = 0.0292$



(c) End of the test $\mu = 1.8893$, $\sigma = 0.0857$

Fig. 1: Histograms for different stages of operation: (a) (b) and (c)(date test : 04/05/2020. fabrication: 09/01/2010)

the test conducted by the company is periodic, with intervals of 30 minutes between each measurement, meaning that before reaching 1.4 V, the battery had at least 1.75 V

Therefore, the equipment used as a solution for the imbalance problem must possess at least four channels to effectively correct the cell imbalance.

III. EXPERIMENTAL VALIDATION ON AN EMULATED SETUP

To improve safety in a controlled environment, an electronic converter was tested to address imbalance issues using an emulated setup with fewer components. We chose an emulated setup for several reasons. In real battery systems, components can degrade during tests, leading to frequent and costly repairs and replacements. Additionally, a controlled environment allows for the analysis of key variables such as individual voltage, current, and temperature in real time.

This experimental study aimed to simulate real-world imbalance in stationary battery banks, show its effects during discharge tests, and validate a device that can individually correct unbalanced components. The methodology had two phases: (1) Testing without intervention and (2) Testing with intervention. Since the main goal is to show the statistical analysis, only the histograms from the intervention test are presented. It is assumed that an imbalance problem would cause the test to fail, which is straightforward.

In this test, since there aren't 60 components in series, battery imbalance doesn't happen naturally. Instead, we forced an imbalance in one component and then performed a spot charge on it using our equipment to fix the problem. We've detailed the test methodology below.

Before each experimental phase, we prepared the battery bank according to current technical standards for testing [2] [3]. Here are the procedures we followed:

1) *Pre-Test Battery Conditioning*: To prepare the battery bank before each experimental phase, we followed current technical standards with these steps:

- 1) Full charging of the battery bank using a constant current source, followed by an equalization step.
- 2) Monitoring of individual element voltages until values stabilized.
- 3) Ensuring all elements were within the manufacturer's recommended operating range.

A. Tests Without Intervention

After preparing the battery bank, a 25% SOC imbalance is introduced into one of the elements:

This procedure created a voltage discrepancy between the affected element and the others, simulating a common scenario in poorly equalized battery banks. As noted in Fig. 2 (a), this element has a lower voltage than the others.

1) *Standard Float Operation*: The battery bank was kept in a float regime for an adequate period. During this time, we monitored the unbalanced element and observed that it did not reach the same voltage level as the others, evidencing the ineffectiveness of passive float in rebalancing the cell. As illustrated in figure ??(a).

2) *Discharge Test*: A controlled current discharge test was conducted, following normative guidelines. The battery bank failed the test, with premature interruption of the discharge due to a sharp voltage drop.

3) *Diagnosing the Cause of Failure*: The unbalanced element was isolated and subjected to an individual test. This demonstrated that it would be capable of sustaining the specified current if adequately charged. It was concluded that the bank's failure was not caused by a physical defect, but rather by the difference in charge between the elements.

B. Active Intervention Test

With the same initial imbalance scenario, a corrective methodology was applied using equipment designed to perform individual charging of the elements. The steps were:

1) *Intervention on the Unbalanced Element*: Upon confirming the imbalance, individual charging of the unbalanced element was performed without removing it from the bank, while the other cells remained in float regime, utilizing the intervention equipment for imbalance correction.

2) *Prolonged Bank Float*: With the complete bank, a new, extended float period was performed, aiming to standardize the voltage across all elements.

3) *Repeat Discharge Test*: The same discharge test was repeated. The battery bank was successful, maintaining the voltage within established limits until the end of the test.

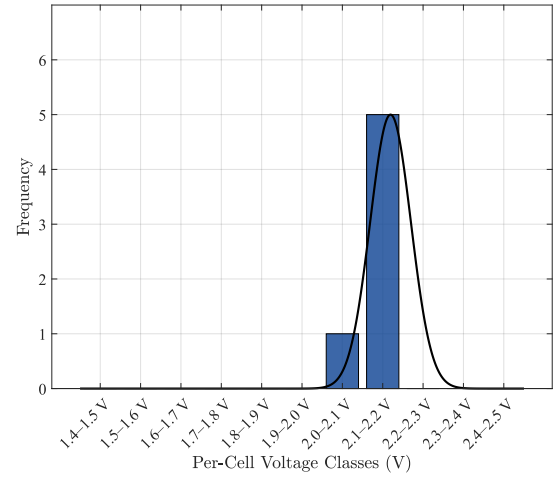
4) *Monitoring the Corrected Element*: During the test, the previously unbalanced element was monitored individually, maintaining performance equivalent to the others, thus proving the effectiveness of the intervention.

IV. CONCLUSIONS

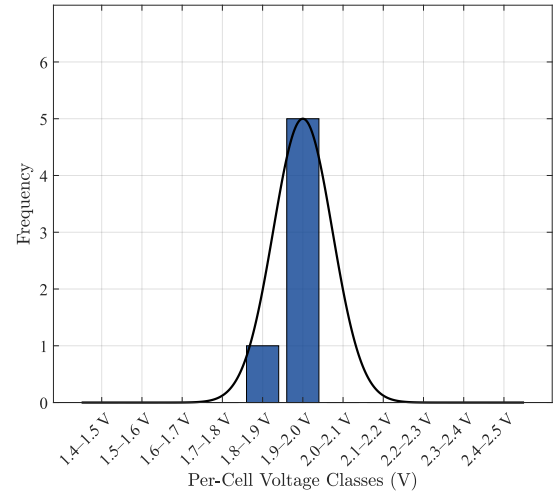
This study addressed the analysis of imbalances in stationary lead-acid battery banks in electrical substations, applying Digital Signal Processing (DSP) techniques and statistical analysis to monitor and assess the operating conditions of these batteries. The main contribution of this work was the identification of imbalance patterns in the battery cells, which can lead to premature degradation of the components and failure of the battery bank.

It was possible to verify that during the discharge tests, elements of the battery bank showed significant voltage deviations, indicating an imbalance that compromises the lifespan of the bank. The application of DSP tools enabled a clear visualization of the imbalance evolution through histograms, in addition to providing essential metrics such as mean and standard deviation, which are crucial for failure diagnosis.

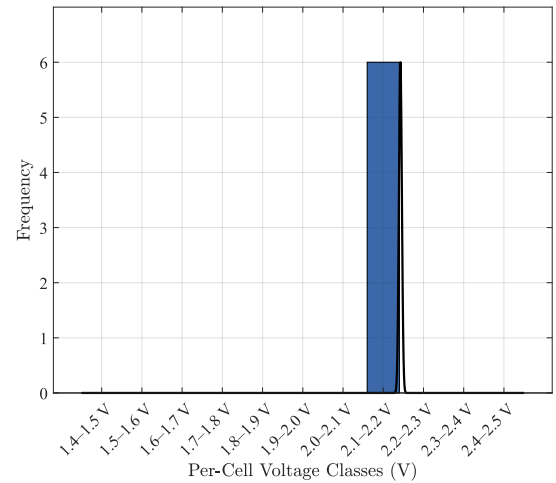
The intervention with a specialized electronic device proved effective in correcting imbalances in the cells, allowing the battery bank to successfully pass the discharge test after individual cell equalization. The use of an emulated setup with fewer components allowed the simulation of realistic scenarios without the costs and risks associated with the use of real batteries.



(a) Float



(b) Test start



(c) After intervention with device

Fig. 2: Histograms for different stages of operation

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